

Answer all questions completely. Calculators may not be used.

Notation: $\mathbb{D} = \{z : |z| < 1\}$, $H(G) = \{f : f \text{ is analytic on the region } G\}$.

1. State and prove the Fundamental Theorem of Algebra.
2. Suppose $f \in H(\mathbb{D})$, f is non-constant and $\lim_{|z| \rightarrow 1} |f(z)| = 1$. Show that $f(\mathbb{D}) \subset \mathbb{D}$.
3. Classify and find the residues at the isolated singularities of the function

$$\frac{1}{z} - \frac{1}{\sin z}.$$

4. Suppose $|a_0| > |a_1| > \cdots > |a_n| > 0$. Show that the polynomial $na_0 + a_1z + a_2z^2 + \cdots a_nz^n$ has no zeros in $\mathbb{D}$.
5. Find all conformal maps from $\mathbb{H} = \{x + iy : y > 0\}$ onto $\mathbb{S} = \{x + iy : |y| < \pi\}$.
6. Let $S_n(z) = \sum_{k=0}^n \frac{z^k}{k!}$, $n = 1, 2, \dots$. Show that for all $r > 0$ there exists $N \in \mathbb{N}$ so that $n \geq N$ and $|z| < r$ implies $S_n(z) \neq 0$.
7. Use the Residue Theorem to evaluate $\int_0^\infty \frac{3x^2 + 1}{(x^2 + 1)(x^2 + 4)} dx$. Justify all steps.
8. Suppose $f : \mathbb{D} \rightarrow \mathbb{D}$ is analytic and has two distinct fixed points (ie, there exist $z_1, z_2 \in \mathbb{D}$, $z_1 \neq z_2$ with $f(z_1) = z_1$ and $f(z_2) = z_2$). Show that f must be the identity map.